VOL. 1 · NO. 22 · DEFEND

REPORTER: _____

The 2026-XJ Daily

Op-ed: what the physics says

HEADLINE · YOUR CLAIM

MY ARGUMENT

I predict ____ based on ____. My model assumes ____. Point to your own work.

THE EVIDENCE BOX

Paste or copy your key calculation and sketch here.

THE LIMITS

The result is limited by ____. A range is not a probability.

PHYSICS vs. DECISION

What can the physics decide? What needs people to decide?

MY REVISION

One step I changed after feedback, and how it changed my result:

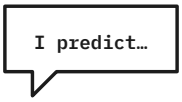
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Talk it through: defend your prediction

Language goal: I can **defend** and **revise** my answer.

DIRECTIONS · Partner A and B read their lines aloud. Listen, then write. Switch each round.

ROUND 1 · SHARE YOUR PREDICTION

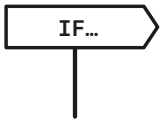


A ASKS "Where will it be, and how sure are you?"

B SAYS "I predict ____ m, with a range of ____ to ____ m."

A CHECKS "Which measurement or graph supports it? Point to it."

ROUND 2 · SWITCH · WHAT DID YOU TREAT AS TRUE?

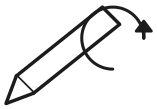


B ASKS "What does your calculation treat as true?"

A SAYS "I treated ____ as true. If that changes, my answer would ____."

B ADDS "I want to add one limit: ____."

ROUND 3 · BOTH · REVISE



A "After listening, I want to change ____ because ____."

B "My prediction is stronger now because ____."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

My final prediction and why…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

DEFEND (1) (2) (3)

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Mic check: drop the Unit 1 verse

Every bar must be true physics. Your partner fact-checks each line.

22 W

MUST INCLUDE

- one motion idea (x, v, a)
- one force idea (ΣF, F = ma)
- a prediction with a range
- a limit of the model

RHYME BANK

force / course · range / change · mass / fast / last · net / set / yet · claim / aim / same · proof / truth

THE VERSE · 8 BARS (v = fact-checked)

BAR	LYRIC (4 beats)	v
1		
2		
3		
4		
5		
6		
7		
8		

PROVE ONE BAR

Pick a bar with a number. Show the math.

FACT-CHECKER

Checked by ____ · bars fixed ____

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Pitch your recommendation

The Tank is the city council. You have 60 seconds and your transfer-task evidence.

22 Z

SLIDE 1 · THE PREDICTION

Central value ____ · range ____ to ____ · model ____

SLIDE 2 · THE EVIDENCE

Your strongest calculation or graph:

SLIDE 3 · THE LIMITS

What the model leaves out:

THE PITCH · 60 SECONDS

"Council members, our physics model predicts ____ under the assumption that ____. The range is ____, so we recommend ____. Physics can tell you ____, but deciding ____ also depends on cost, feasibility, and priorities."

SHARK QUESTION

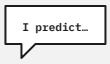
"How sure are you?" Your answer – without calling the range a probability:

DEFEND 1 2 3

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Say it, then write it: defend it

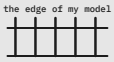
Language goal: I can **defend** a claim.



claim
afirmación · afirmação
what I say is true



evidence
evidencia · evidência
numbers, graphs, sketches



limit
límite · limite
what the model leaves out



revise
revisar · revisar
fix and improve

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: claim · evidence · reasoning · limit · revise

My ____ is the prediction. My ____ is my calculation and sketch. My ____ explains why they connect.

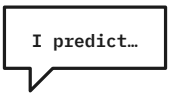
Every model has a ____. After feedback, I ____ my work.

2 · SAY IT TO A PARTNER, THEN WRITE IT



I predict ____ based on ____ .
My model assumes ____, and it is limited by ____ .

3 · SOLVE WITH THE STEPS



Your transfer task: state your prediction and range.

STEP 1 · CLAIM	STEP 2 · EVIDENCE	STEP 3 · LIMIT
"I predict ____."	"Look at my ____."	"It leaves out ____."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ At first I thought ____. Now my evidence shows ____ .

☐ A skill I can use on my own is ____ .

☐ I still need practice with ____ .

I said it ☐ · I wrote it ☐

DEFEND 1 2 3

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Peer review: settle the defense

22 I

DIRECTIONS · Classmates disagree about how to defend a prediction. Who is right?

ROUND 1

ANA
"My number is right, so I don't need to show work."

BEN
"A correct-looking number doesn't show your reasoning. Point to the evidence."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 2

KAI
"Our 5 ± 1 m means there's a 50% chance of 5 m."

LU
"It's a bound from our errors, not a probability."

Who's right? _____ · Evidence: _____

Fixed claim: _____

ROUND 3

MAYA
"If my partner finds a mistake, I just copy their answer."

RAJ
"Show the corrected step yourself and explain how it changes later results."

Who's right? _____ · Evidence: _____

Fixed claim: _____

YOUR TURN · WRITE A TRAP

Write a claim about today's idea that sounds right but is wrong. Trade and settle it.

DEFEND 1 2 3

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Comic: will 2026-XJ hit us?

22 CS

YOUR READER
A curious 10-year-old who asks, "Will the asteroid hit us?"

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◌ **BLUE** — Position data with units
- ☐ ◌ **PURPLE** — The asteroid's velocity arrow
- ☐ ◯ **PENCIL** — One sentence on how sure we are, and why

1 · THE EVIDENCE *Show what the team measured.*

CAPTION _____

2 · THE REASONING *Show the physics that connects the evidence to a prediction.*

CAPTION _____

3 · THE ANSWER *Give your recommendation and how sure you are.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **evidence · prediction · uncertainty · recommendation**

_____

_____

DEFEND 1 2 3

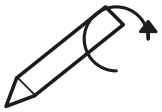
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Doodle Notes · revision

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Revision · revisión (ES) · revisão (PT)



TODAY'S CASE

A team predicts a cart will finish at 6–8 m. The next measured position is 9 m.

01 · SKETCH AND LABEL

Draw the predicted range and measured dot on the same blue position axis.

02 · CHANGE THE PICTURE

Label the discrepancy. Sketch one measurement you would check or collect before revising the model.

Annotate your sketches. Circle the mark that best shows revision. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

I would revise ___ if the new evidence shows …

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Frayer Model · revision

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Revision · revisión (ES) · revisão (PT)

A · A team checks new measurements and changes its model assumption.

B · A team erases the point because it falls outside the forecast.

01 · DEFINITION

What does revision mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

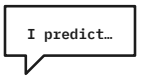
I chose card ___ as my example because …

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Picture Dictionary · revision

I can connect each word to a picture and use it to explain this case.

Name: _____ My home language: _____

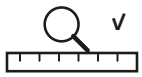


claim

afirmación · afirmação

what I say is true

My word or label: _____

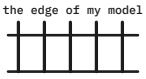


evidence

evidencia · evidência

numbers, graphs, sketches

My word or label: _____



limit

límite · limite

what the model leaves out

My word or label: _____

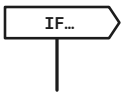


revise

revisar · revisar

fix and improve

My word or label: _____

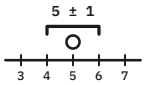


assumption

supuesto · hipótese

a condition you take as true in the model

My word or label: _____



range

intervalo · intervalo

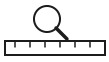
the span of possible predictions

My word or label: _____

MATCH · WRITE THE ENGLISH WORD UNDER EACH PICTURE



_____



_____



_____

Use two words to explain the case: A team predicts a cart will finish at 6–8 m. The next measured position is 9 m.

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